COMSATS University Islamabad (Lahore Campus)

Department of Computer Science

Quiz # 2 — Fall 2026

Course Title:	Web Technologies (CSC336)	**Total Marks:**	10 Marks
Coverage / Focus:	CLO-2	**Time Allowed:**	25 Minutes
Student Name:		**Reg. No:**	

Instructions: Answer both questions clearly with precise technical explanations and exact syntax.

Question 1: CLO-2 [6 Marks]

A developer designs a responsive website and defines multiple CSS media queries (e.g., @media (max-width: 600px) { ... }) to reorganize elements for smartphone users. However, when tested on an actual mobile device (physical screen width 390px, device-pixel-ratio = 3), the browser continues to render the full desktop layout scaled down, completely ignoring the mobile media queries.

(a) Explain the fundamental reason why the mobile browser fails to trigger the mobile media query despite the screen width being 390px. Distinguish clearly between the visual viewport and the layout viewport in your explanation. **[2 Marks]**

(b) Write the exact HTML <meta> tag that must be placed in the <head> of the document to resolve this issue. **[2 Marks]**

(c) Explain specifically what each property-value pair inside the content attribute of your tag instructs the browser engine to do. **[2 Marks]**

Question 2: CLO-2 [Client-Server Communication: GET vs. POST via Fetch API] [5 Marks]

A student records portal provides two RESTful endpoints:

- GET https://api.portal.edu/api/students?dept=CS (to retrieve student records by department)
- POST https://api.portal.edu/api/students (to register a new student record)

(a) Write the JavaScript code using fetch() to execute both operations:

i. Fetch and print the list of CS students using the GET endpoint.

ii. Register a new student using the POST endpoint with the object:
const newStudent = {name: "Sarah Khan", rollNo: "SP23-BCS-042", cgpa: 3.75 };
Ensure the request properly sets the method, header(s), and serialized body. **[2 Marks]**

(b) Compare how data is transmitted between GET and POST in terms of:

1. Data placement (URL vs. Request Body)
2. Browser caching and URL visibility (security considerations)
3. Why a GET request does not typically require a 'Content-Type' header, whereas a POST sending JSON does. **[2 Marks]**